Problem 7: Gated DeltaNet

GStar Bootcamp

Abstract

Combine scalar forgetting with delta correction in recurrent and chunked Triton kernels.

Starter file: `problem_7.py`. **Checker:** `python autograder.py -p7`. **Reading path:** Overview, P5, P6. Next: P8.

Mathematical background: forgetting and correction solve different problems

Delta correction leaves key-orthogonal directions unchanged; forgetting can also remove information in those directions. Forgetting plus an additive write does not correct the stored prediction. Gated DeltaNet combines both operations [1].

Let $\alpha_t = e^{g_t}$. Its homogeneous transition is

$$\mathbf{S}_t = \alpha_t(\mathbf{I} - \beta_t \boldsymbol{k}_t \boldsymbol{k}_t^\top)\mathbf{S}_{t-1} + \beta_t \boldsymbol{k}_t \boldsymbol{v}_t^\top.$$

For a unit key, the transition has eigenvalue $\alpha_t(1-\beta_t)$ along that key and eigenvalue α_t in orthogonal directions. Thus β_t controls correction of the keyed association, while α_t controls global retention. Setting $\beta_t = 0$ leaves pure forgetting; setting $g_t = 0$ recovers DeltaNet.

Scalar decay commutes with the delta transition, so relative decay can multiply an ordinary dot product. The new write still has coefficient β_t, not $\alpha_t\beta_t$. Prefix differences transport earlier corrections without decaying the current one. P8 replaces scalar decay with a diagonal gate, which generally does not commute.

Before coding. Consider one key-aligned component and one orthogonal component. State how each changes with $\beta_t = 1$ and $0 < \alpha_t < 1$ before considering the new write.

Combine scalar forgetting and delta correction

Gated DeltaNet applies a scalar decay to the previous state before computing the residual [1]:

$$\overline{\mathbf{S}}_t = e^{g_t}\mathbf{S}_{t-1}, \tag{7.1}$$

$$\boldsymbol{u}_t = \beta_t(\boldsymbol{v}_t - \overline{\mathbf{S}}_t^\top \boldsymbol{k}_t), \tag{7.2}$$

$$\mathbf{S}_t = \overline{\mathbf{S}}_t + \boldsymbol{k}_t \boldsymbol{u}_t^\top, \quad \boldsymbol{o}_t = \mathbf{S}_t^\top \boldsymbol{q}_t. \tag{7.3}$$

Equivalently, $\mathbf{S}_t = e^{g_t}(\mathbf{I} - \beta_t \boldsymbol{k}_t \boldsymbol{k}_t^\top)\mathbf{S}_{t-1} + \beta_t \boldsymbol{k}_t \boldsymbol{v}_t^\top$. The current write is not multiplied by the current forget gate.

Scalar gated WY

Let $b_r = \sum_{j=1}^{r} g_j$ within a chunk. Define

$$L_{ru} = \begin{cases} \beta_r(\boldsymbol{k}_r^\top \boldsymbol{k}_u) e^{b_r - b_u}, & u < r, \\ 0, & u \geq r, \end{cases} \tag{7.4}$$

$$\mathbf{A} = (\mathbf{I} + \mathbf{L})^{-1}, \tag{7.5}$$

$$\mathbf{W} = \mathbf{A}\mathbf{D}_\beta \mathbf{K}^{\text{start}}, \quad \mathbf{U} = \mathbf{A}\mathbf{D}_\beta \mathbf{V}, \quad \mathbf{P} = \mathbf{U} - \mathbf{W}\mathbf{S}_{\text{in}}, \tag{7.6}$$

where $K_{rd}^{\text{start}} = e^{b_r} K_{rd}$. Exit-decayed keys use $K_{ud}^{\text{end}} = e^{b_C - b_u} K_{ud}$, and entry-decayed queries use $Q_{rd}^{\text{start}} = e^{b_r} Q_{rd}$.

$$\mathbf{S}_{\text{out}} = e^{b_C} \mathbf{S}_{\text{in}} + (\mathbf{K}^{\text{end}})^\top \mathbf{P}, \tag{7.7}$$

$$\mathbf{O} = \left[\mathbf{Q}^{\text{start}} \mathbf{S}_{\text{in}} + (\text{tril}(\mathbf{Q}\mathbf{K}^\top) \odot \mathbf{D})\mathbf{P}\right], \tag{7.8}$$

with $D_{ru} = e^{b_r - b_u}$ on the inclusive lower triangle and zero elsewhere.

Derivation hint

Expand the decayed entry state and each earlier correction up to local row r. Earlier writes acquire relative decay $e^{b_r - b_u}$. Substitute that expansion into the residual, as you did in Problem 6. Scalar decay factors leave the dot product, producing the simple pairwise factor above.

Problem 7: implementation steps and checks

1. **Recurrent forgetting (20 points).** Decay the previous state; predict from the decayed state; write the weighted error; read the updated state.
2. **Scalar gated WY (30 points).** Compute chunk-local scalar log-prefixes. Construct strictly lower gate differences safely before exponentiation. Weight rows by β and reuse the triangular primitive from Problem 6.
3. **State and output (40 points).** Transform keys toward the chunk start for $\mathbf{W}$, toward the chunk end for the state write, and queries toward their local positions for the entry-state read. Complete Eqs. (7.7)–(7.8).
4. **Explain update ordering (10 points).** Explain why decaying the already-updated state would also decay the current write and therefore compute a different recurrence.

```
# Recurrent Gated DeltaNet
# TODO P7.A: scalar state decay BEFORE residual prediction.
# Reuse your residual update and output read from Problem 5.

# Gated WY construction
# TODO P7.B: prefix differences, strict mask, safe exponentiation.
# TODO P7.C: use entry-decayed keys in W; construct U as before.

# Chunk state/output
# TODO P7.D: exit-decayed keys and decayed entry state.
# TODO P7.E: entry-decayed queries and inclusive score decay.
```

Worked ordering check

In one dimension let $S_{t-1} = 2$, $e^{g_t} = 1/2$, $k_t = 1$, $v_t = 5$, and $\beta_t = 1$. Correct execution gives $\overline{S}_t = 1$, residual 4, and $S_t = 5$. A delta update followed by decay would first produce 5 and then incorrectly reduce it to 2.5.

Verification

- $g = 0$ must recover DeltaNet in both forms.

- $\beta = 0$ must retain only gate decay and state reads.
- With $\beta = 1$ and repeated unit keys, keyed predictions should follow the current value.
- Strong gates must produce finite outputs and states.
- A nonzero initial state and a sequence split must agree with full recurrent execution within the calibrated numerical tolerance.

Use a safe exponent argument for masked entries. Forming the full upper-triangle gate differences and exponentiating before masking can overflow even though only lower-triangle entries are needed.

Problem 7: derive scalar-gated corrections explicitly

The state immediately before the local correction

Before writing correction $\boldsymbol{u}_r$, the gated recurrence has

$$\overline{\mathbf{S}}_r = e^{b_r}\mathbf{S}_{\text{in}} + \sum_{j<r} e^{b_r - b_j}\boldsymbol{k}_j\boldsymbol{u}_j^\top.$$

Substitute this into $\boldsymbol{u}_r = \beta_r(\boldsymbol{v}_r - \overline{\mathbf{S}}_r^\top \boldsymbol{k}_r)$:

$$\boldsymbol{u}_r = \beta_r \left[\boldsymbol{v}_r - e^{b_r}\mathbf{S}_{\text{in}}^\top \boldsymbol{k}_r - \sum_{j<r} e^{b_r - b_j}(\boldsymbol{k}_r^\top \boldsymbol{k}_j)\boldsymbol{u}_j \right].$$

The scalar decay can leave the channel reduction. It modifies the strict coefficient matrix and transforms the entry-state key read, while the triangular-system structure from P6 remains unchanged.

A two-token example with forgetting

Use unit scalar keys, beta one, values $(2, 5)$, entry memory 1, and decays $(1/2, 1/4)$. The first decayed memory is $1/2$, so $u_1 = 3/2$ and $S_1 = 2$. The second decayed memory is $1/2$, so $u_2 = 9/2$ and $S_2 = 5$.

The prefix decays are $e^{b_1} = 1/2$ and $e^{b_2} = 1/8$. The relative coefficient from token 1 to token 2 is $1/4$, giving

$$\mathbf{L} = \begin{bmatrix} 0 & 0 \\ 1/4 & 0 \end{bmatrix}, \quad \mathbf{A} = \begin{bmatrix} 1 & 0 \\ -1/4 & 1 \end{bmatrix}, \quad \mathbf{W} = \begin{bmatrix} 1/2 \\ 0 \end{bmatrix}, \quad \mathbf{U} = \begin{bmatrix} 2 \\ 9/2 \end{bmatrix}.$$

Then $\mathbf{P} = (3/2, 9/2)^\top$ and

$$S_{\text{out}} = \tfrac{1}{8}S_{\text{in}} + \tfrac{1}{4}P_1 + P_2 = 5.$$

The current correction has exit weight one. The earlier correction has the second token's gate, not the product of both gates.

Mask before exponentiation

Valid earlier positions satisfy $b_r - b_j \le 0$. Invalid differences can be positive and large: replace them with a safe exponent argument before exponentiation, then zero their scores. Masking an overflowed infinity afterward can produce NaN.

Scalar decay can commute; channel decay generally cannot

Scalar decay commutes with the delta transition, allowing an ordinary dot product times a pairwise decay. P8 uses a diagonal gate, whose placement relative to the rank-one transition must be preserved.

References

[1] S. Yang, J. Kautz, and A. Hatamizadeh. *Gated Delta Networks: Improving Mamba2 with Delta Rule*. 2024/2025.
https://arxiv.org/abs/2412.06464